

# When Does Intermediate Reward Actually Help?

Outcome-only credit assignment vs. explicit progress supervision  
for multi-turn LLM search agents — Research Proposal with Preliminary  
Results

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## Abstract

Reinforcement learning for multi-turn LLM agents almost always optimizes a *sparse outcome reward*: one scalar at the end of a trajectory. Whether an agent benefits from an additional *intermediate reward signal* — supplied by an explicit reward model, a privileged verifiable heuristic, or recovered implicitly by the credit-assignment (CA) machinery of the RL algorithm itself — remains unresolved, and published comparisons rarely control for reward density, compute, or seeds. We propose a controlled study on the Search-R1 question-answering task: a Qwen3-1.7B agent that interleaves reasoning, search-engine calls, and a final answer over up to 4 turns (8 in a stress test). We compare seven credit-assignment arms spanning implicit reward decomposition (token- and turn-level critics; group-relative estimators GRPO/GiGPO/HCAPO) against explicit progress supervision (a privileged answer-exposure step reward, B1) and a density-matched shuffled control (B1-shuffle) that uniquely separates the reward’s *information content* from its *optimization effects*. Preliminary results (Wave 0–1, 9 completed or near-complete 500-step runs) show: (i) all arms clear the untrained baseline (0.225 macro-EM) by a wide margin, with outcome-only group-relative methods leading (GiGPO  $\approx 0.41$  val macro-EM, GRPO 0.391 val / 0.395 on the full 51,713-question test set); (ii) explicit progress reward so far adds *no* value over its shuffled control — the control currently *beats* it on matched seeds — consistent with a reward-density rather than content effect; and (iii) a reproducible late-training instability (response-truncation runaway) afflicts the critic-based arms, most severely the content-bearing B1 arm, while both shuffle runs stay clean. We outline the pre-registered analyses (third seeds, 8-turn horizon, per-dataset multi-hop subgroup, and a Monte-Carlo credit-alignment diagnostic) that will settle the hypotheses.

## 1 Motivation

Agentic RL fine-tuning of LLMs — tool use, web research, multi-step reasoning — is bottlenecked by **credit assignment**: the outcome reward arrives only after a sequence of interdependent decisions. The community’s responses fall into two families that are rarely compared under matched conditions:

1. **Implicit reward models.** The RL algorithm itself decomposes the outcome reward: a learned value network (token-level or turn-level critic) is, functionally, an implicit per-step reward model; group-relative estimators (GRPO and its turn-aware descendants GiGPO, HCAPO) sidestep the critic but still implicitly assign per-step credit through grouped baselines and anchor-state matching.
2. **Explicit intermediate rewards.** A process/progress reward is injected at each step — from a trained process reward model (PRM), a prompted LLM judge, or a verifiable domain heuristic.

The central question of this proposal is **when the second family adds value beyond the first**, and whether apparent gains from intermediate rewards survive a control that matches

their density and magnitude but destroys their timing/content. This control — rarely run in the literature — is the experiment’s key instrument.

## 2 Hypotheses

- **H1 (granularity).** Turn-level credit (one credit unit per agent action) is a better-matched inductive bias than token-level credit for multi-turn agents; it should not underperform token-level PPO at matched compute.
- **H2 (implicit recovery).** Outcome-only group-relative CA can recover useful intermediate credit without any explicit step reward; measured not only by final accuracy but by a quantitative credit-alignment diagnostic (assigned credit vs. Monte-Carlo continuation value, Sec. 5).
- **H3 (content vs. density — the core hypothesis).** If explicit progress supervision helps through its *information content*, a privileged step reward (B1) must beat a density-matched shuffled control (B1-shuffle). If  $B1 \approx B1\text{-shuffle} > B0$  (sparse), the benefit is an optimization/density effect that any well-shaped dense signal — including a contentless one — provides. Preliminary reading: the latter.
- **H4 (horizon scaling).** Any content benefit should grow with task horizon and difficulty (multi-hop questions; 8-turn budget), where progress information has more room to discriminate.
- **H5 (scalable evaluator, staged).** A prompted step evaluator can approximate the privileged signal only if H3 shows content matters; H5 is gated on H3’s outcome.

## 3 Experimental design

### 3.1 Task, dataset, and reward

**Task.** Search-R1 protocol: the agent answers open-domain questions by interleaving `<think>` reasoning with either a `<search>query</search>` call to a retriever or a final `<answer>...</answer>`, for up to 4 turns (main protocol) or 8 turns (stress test). Retrieval is dense e5 over the 2018 Wikipedia dump (wiki-18, 61 GB FAISS index), top-3 passages per query, served by a co-located retrieval server.

**Training data.** The standard Search-R1 training mixture — Natural Questions + HotpotQA training splits (HF `PeterJinGo/nq_hotpotqa_train`, ~170k questions) — preprocessed by the versioned pipeline in the repository. Validation during training uses a fixed stratified 2,048-question subset (`val_2048`); final evaluation is the full paper-protocol test set: **51,713 questions over 7 datasets** (NQ, TriviaQA, PopQA single-hop; HotpotQA, 2WikiMultihopQA, MuSiQue, Bamboo multi-hop), greedy decoding, exact match (EM).

**Outcome reward.** Rule-based EM on the final answer (1/0), with a small invalid-action penalty. No learned reward model anywhere in the core tier — by design, so that the *only* intermediate signal is the one we inject and control.

**Model.** Qwen3-1.7B (thinking mode, 2,048-token responses, 4,096-token prompts), identical across all arms; batch 256, group size 5, 500 PPO steps, identical LR/KL settings. A pre-registered feasibility gate on the *untrained* model (nonzero EM floor, valid-action rate  $\geq 90\%$ , search-invocation rate  $\geq 50\%$ , reward variance within rollout groups) passed at 1.7B, so the fleet stays at this scale (fallback: Qwen3-4B).

| Arm         | Estimator                         | Critic | Where the “reward model” lives                                                                                                                       |
|-------------|-----------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| TOKEN-PPO   | GAE (token)                       | yes    | implicit: token-level value network                                                                                                                  |
| TURN-PPO/B0 | GAE (turn)                        | yes    | implicit: turn-level value network                                                                                                                   |
| TOKEN-GRPO  | GRPO                              | no     | implicit: group-relative outcome baseline                                                                                                            |
| GiGPO       | GiGPO                             | no     | implicit: episode + anchor-state grouped baselines                                                                                                   |
| HCAPO       | HCAPO                             | no     | implicit: hindsight-conditioned advantage (Wave 2)                                                                                                   |
| B1          | GAE (turn) + step reward          | yes    | <b>explicit:</b> privileged answer-exposure signal, +0.2 the first turn whose retrieved passages contain the gold answer                             |
| B1-SHUFFLE  | GAE (turn) + shuffled step reward | yes    | <b>explicit, contentless:</b> B1’s rewards re-assigned uniformly to another non-terminal search turn — matched count and magnitude, destroyed timing |

**Table 1:** Core arms. B1 is deliberately *privileged and verifiable* (it peeks at the gold answer) — an upper-bound proxy for a perfect progress reward model, free of judge noise. The shuffled control isolates information content from reward density. A prompted-judge arm (B2-lite) is staged behind H3.

### 3.2 Conditions: a reward-model taxonomy

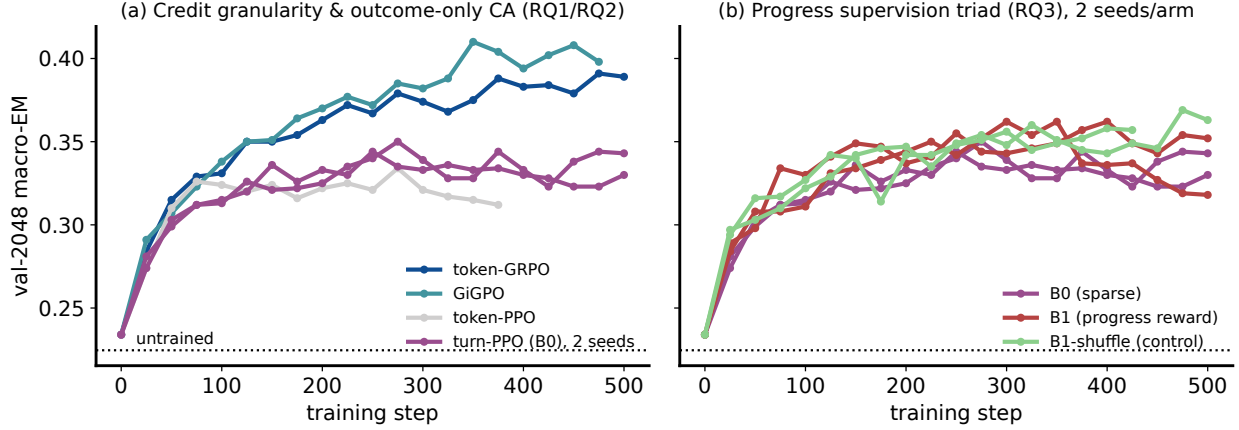
**Pre-registered decision rules** (fixed before any curve was seen): primary comparison = fixed-budget final checkpoint; sample-efficiency = validation AUC; best-val checkpoint reported but never the headline. RQ3 is judged on per-seed final-EM and val-AUC deltas with seed ranges (2–3 seeds per arm), plus a paired per-dataset breakdown with the multi-hop subgroup pre-registered as the mechanism readout. Interpretation grid:  $B1 > \text{shuffle} \approx B0 \Rightarrow \text{content matters}$ ;  $B1 \approx \text{shuffle} > B0 \Rightarrow \text{density effect}$ ; all  $\approx \Rightarrow$  this signal does not help at this horizon.

## 4 Preliminary results

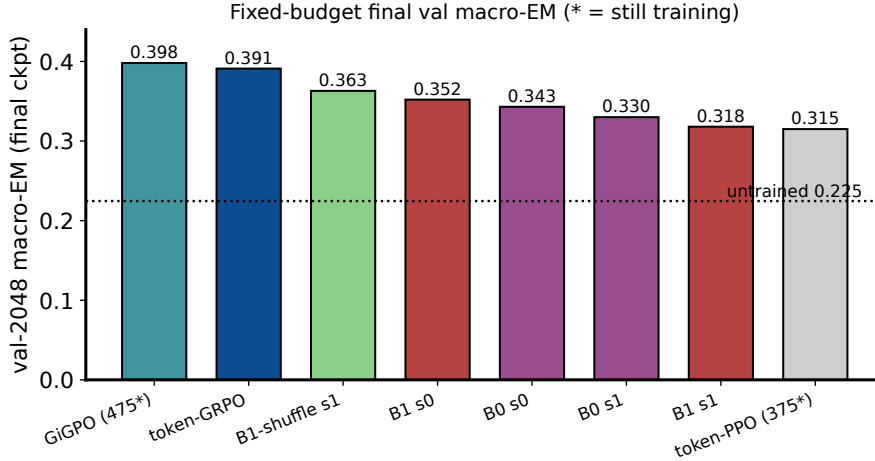
Nine 500-step runs are complete or near-complete (Wave 0–1 + partial Wave 2); three third-seed runs and the 8-turn stress test are in flight. Untrained baseline: 0.225 val macro-EM.

### 4.1 Interim readings

1. **H1 (granularity):** supported so far — turn-PPO (0.343/0.330)  $\geq$  token-PPO ( $\sim$ 0.315, still training) at matched config, though both trail the critic-free arms.
2. **H2 (implicit recovery):** strongly supported on accuracy — GiGPO and GRPO lead the table without any intermediate reward. The credit-alignment diagnostic will test whether they also assign *correct* intermediate credit.
3. **H3 (content vs. density):** at 2/2/2 seeds, the pre-registered step-150 window readout gave shuffle–B0 positive on both seeds and B1–shuffle straddling zero; at final checkpoints B1-shuffle (0.363) currently *beats* B1 (0.352/0.318) while suffering none of its instability. Interim call: **reward-density/optimization effect, not progress-content supervision** — pending third seeds and val-AUC.
4. **Stability as a first-class finding:** the truncation tripwire (Fig. 4) reproduces across 4/4 critic arms and never in the 3 clean shuffle/GiGPO runs monitored at the same wall-clock; the formal report will treat late-training length instability as a cost of this class of explicit step reward.



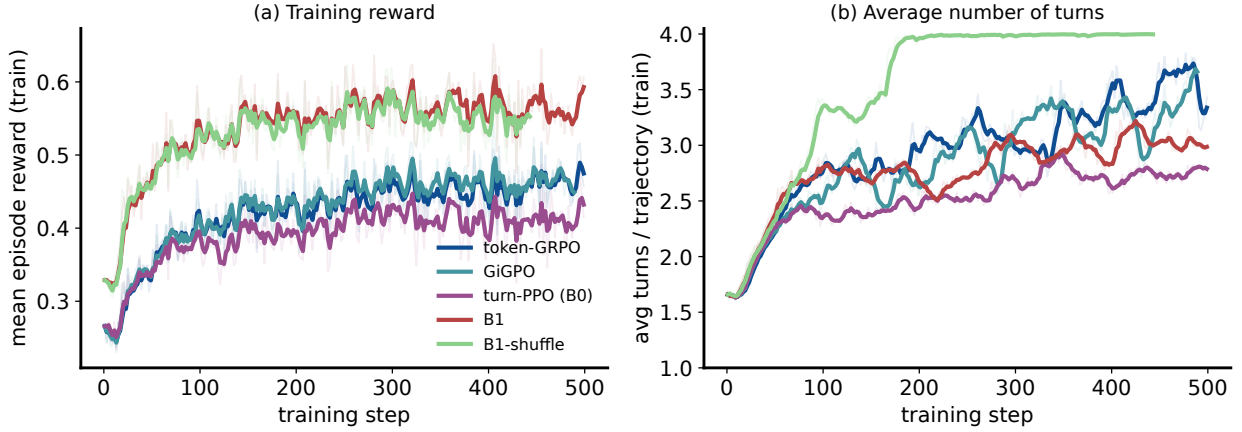
**Figure 1:** Validation macro-EM (2,048 fixed questions, greedy, every 25 steps). **(a)** Outcome-only group-relative methods lead: GiGPO (still training, best 0.410) tracks above token-GRPO (final 0.391); both critic-based PPO arms plateau near 0.33–0.34; token-PPO is weakest. **(b)** The RQ3 triad at two seeds per arm: the three conditions are statistically inseparable for most of training; late-run divergence is driven by the instability in Fig. 4.



**Figure 2:** Fixed-budget final val macro-EM (the pre-registered primary comparison). The wave-0 GRPO checkpoint scores **0.3951 macro-EM on the full 51,713-question test set** (0.4993 single-hop / 0.3170 multi-hop), within +0.004 of its val-2048 proxy — validating the proxy. Seed pairs: B0 0.343/0.330, B1 0.352/0.318, B1-shuffle 0.363/(s0 running at 0.358).

## 5 Remaining work

(i) Third seeds for the RQ3 triad (in flight) to tighten the pre-declared seed ranges; (ii) the **8-turn horizon stress test** (required; first run launched) — H4’s direct test, where shuffle has more eligible turns and content has more room to matter; (iii) HCAPO, GiGPO/GRPO second seeds (wrappers staged); (iv) the **credit-alignment diagnostic**: for logged trajectories, estimate each prefix’s Monte-Carlo continuation value  $\hat{V}(s_t)$  by resuming from snapshots under the same checkpoint, and compare each method’s assigned turn credit against  $\Delta\hat{V}_t = \hat{V}(s_{t+1}) - \hat{V}(s_t)$  (pooled Spearman, bootstrap CIs) — upgrading H2 from a benchmark claim to a measurement of credit fidelity; (v) full-set evals of all pre-registered final checkpoints; (vi) B2-lite (prompted step evaluator) only if the third seeds overturn the density reading.



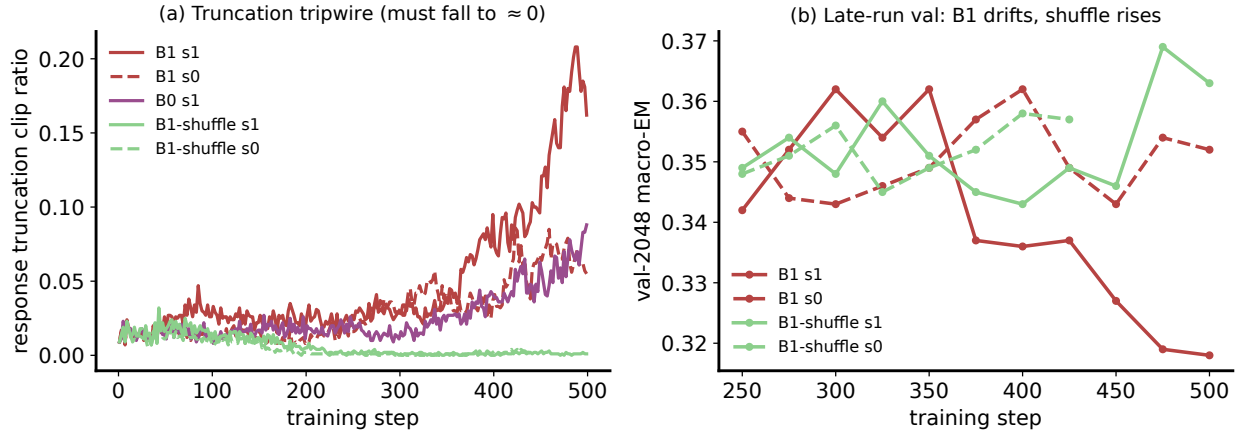
**Figure 3:** Training dynamics (seed-0 arms; EMA-smoothed). **(a)** Mean episode reward. B1/B1-shuffle sit higher by construction (their reward includes the +0.2 step bonus). **(b)** Average turns per trajectory grows from  $\sim 1.7$  to  $\sim 3$  across all arms — agents learn to search more before answering. Notably, the *shuffled* step-reward arm saturates the 4-turn budget entirely: a contentless dense reward on search turns teaches “search more,” not “search better” — yet loses nothing in accuracy, foreshadowing the density reading of H3.

## A Example trajectory (trained GRPO agent, full-set eval, EM = 1)

|                                                                                           |                                                                                                                                                                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| QUESTION: Who delivered their famous 'The Lady's Not For Turning' speech in October 1980? |                                                                                                                                                                                                                                                                                                                                                                 |
| TURN 1                                                                                    | <code>&lt;search&gt;who delivered their famous 'The Lady's Not For Turning' speech in October 1980&lt;/search&gt;</code><br><code>&lt;information&gt;Doc 1: "The lady's not for turning" ... was a phrase used by Margaret Thatcher, then Prime Minister, in her speech to the Conservative Party Conference on 10 October 1980. ...&lt;/information&gt;</code> |
| TURN 2                                                                                    | <code>&lt;search&gt;who delivered the 'The Lady's Not For Turning' speech in 1980&lt;/search&gt;</code><br><code>&lt;information&gt;Doc 1: ... Margaret Thatcher, then Prime Minister ... (confirmatory retrieval)&lt;/information&gt;</code>                                                                                                                   |
| TURN 3                                                                                    | <code>&lt;think&gt;... Doc 1 and Doc 2 mention Margaret Thatcher. She was the Prime Minister at that time. The speech was delivered at the Conservative Party Conference on October 10, 1980. ... I don't see any conflicting information.&lt;/think&gt;</code><br><code>&lt;answer&gt;Margaret Thatcher&lt;/answer&gt;</code> --> outcome reward 1.0           |

Under B1, turn 1 would additionally receive the +0.2 answer-exposure step reward (first retrieval containing “Margaret Thatcher”); under B1-SHUFFLE that same +0.2 would be re-assigned uniformly among turns 1–2. Redundant confirmatory searches as in turn 2 are exactly the behavior the average-turns saturation of Fig. 3b amplifies.

**Reproducibility.** Figures: `assets/extract_metrics.py` + `assets/build_figures.py` (CSV per run under `assets/csv/`); training logs, ledger, and the append-only decision log live in the project repository (Agentic-RL-CA, branch `agentic-rl-ca`); pre-registration in `docs/plan.md`.



**Figure 4:** The main preliminary *mechanistic* finding. **(a)** Response-truncation rate (fraction of turns hitting the 2,048-token cap; must fall to  $\approx 0$  in a healthy run). Every critic-based arm drifts upward late in training; the content-bearing B1 seed-1 run runs away (0.21 by step 500) and its validation accuracy collapses in lockstep, **(b)**. Both B1-shuffle seeds stay at  $\approx 0$  truncation and *rise* late. The explicit progress reward is implicated in destabilizing generation length; the density-matched control is not.